ASYMPTOTICALLY SHARP LINE COVERS OF REFLEXIVE FINITE-FIELD PLANE CURVES

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Editorial note. Complete original mathematical body and embedded bibliography retained. Authors own mathematical correctness; no proof repair, added assumptions, independent counterexample search or certification.

Editorial note. The explicit nonreflexive curve is the original authors construction and proof, not a new obstruction supplied during collection. Its p,q,n and degree hypotheses retained exactly.

Editorial note. Reflexivity is required only for the geometrically irreducible case as stated; the original finite-field-irreducible but geometrically reducible argument remains context.

Editorial note. Main outer proof is the authors native section narrative; no artificial proof environment added. Direct large-characteristic and numerical consequences not separately counted.

Allowed standard prerequisites (editorial).

- Finite-field plane curves, reflexivity, projection monodromy and nearly regular hypergraph covers. Native geometric versus finite-field irreducibility distinctions, separable Gauss-map definition, Chebotarev and Pippenger-Spencer statements and all projection incidence arguments retained.. Original full mathematical body and bibliography retained; the large-characteristic corollary remains context, and the explicit nonreflexive family is the authors separate constructive result.

1. Introduction

Let $C \subset \mathbb{P}^2$ be an $\mathbb{F}_q$-irreducible projective plane curve of degree $d \geq 2$ defined over $\mathbb{F}_q$. We are interested in the following natural question.

Question 1.1. What is the minimum number of $\mathbb{F}_q$-lines in $\mathbb{P}^2$ whose union contains all points of $C(\mathbb{F}_q)$?

We write $\mathrm{cov}(C)$ for this minimum. If C is not geometrically irreducible, then the problem is easy. Indeed, over $\overline{\mathbb{F}_q}$, the curve C is the union of $r \geq 2$ Galois-conjugate irreducible components $C_1, \ldots, C_r$, each of degree d/r. Every point of $C(\mathbb{F}_q)$ lies on each of these components. In particular, $C(\mathbb{F}_q) \subset C_1 \cap C_2$, and Bézout's theorem gives

$$|C(\mathbb{F}_q)| \leq \left(\frac{d}{r}\right)^2 \leq \frac{d^2}{4}.$$

Consequently, $C(\mathbb{F}_q)$ can be covered by $O(d^2)$ $\mathbb{F}_q$-lines. We may therefore focus on the geometrically irreducible case.

By pairing up the points, we can always cover $C(\mathbb{F}_q)$ using $q/2 + O_d(\sqrt{q})$ lines. However, as we will explain below, we can do much better.

Bézout's theorem shows that each line contains at most d distinct $\mathbb{F}_q$-points of C. By the Aubry–Perret bound [AP96] for geometrically irreducible plane curves, we have $|C(\mathbb{F}_q)| \geq q + 1 - (d-1)(d-2)\sqrt{q}$. It follows that we need at least

$$\frac{q+1}{d} - \frac{(d-1)(d-2)\sqrt{q}}{d} \tag{1}$$

$\mathbb{F}_q$-lines to cover $C(\mathbb{F}_q)$.

We write C_{sm} for the smooth locus of C. Given a geometrically irreducible plane curve C, the *dual curve* $C^\vee \subset (\mathbb{P}^2)^\vee$ parametrizes the tangent lines to C. We say that C is *reflexive* if its Gauss map

$$C_{\mathrm{sm}} \longrightarrow C^\vee, \qquad P \longmapsto T_P C,$$

is separable.

Our main result is the following theorem.

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Theorem 1.2. *Fix $d \geq 2$ and $\varepsilon > 0$. There exists a constant $M_{d,\varepsilon}$ such that the following holds. Let $C \subset \mathbb{P}^2$ be an $\mathbb{F}_q$-irreducible plane curve of degree d defined over $\mathbb{F}_q$. If C is geometrically irreducible, assume in addition that C is reflexive. If $q > M_{d,\varepsilon}$, then $C(\mathbb{F}_q)$ can be covered by at most $\left(\frac{1}{d} + \varepsilon\right) q$ many $\mathbb{F}_q$-lines.*

Corollary 1.3. *The conclusion of Theorem 1.2 holds for all $\mathbb{F}_q$-irreducible curves C whenever* $\operatorname{char}(\mathbb{F}_q) > d$.

Proof. If C is not geometrically irreducible, then Theorem 1.2 applies without any further assumption. Suppose, therefore, that C is geometrically irreducible. A nonreflexive plane curve that is not a line has degree at least $\operatorname{char}(\mathbb{F}_q)$; see [Hef89, p. 5]. Since C is not a line and has degree $d < \operatorname{char}(\mathbb{F}_q)$, it must be reflexive. The conclusion now follows from Theorem 1.2. □

In view of the lower bound (1), Theorem 1.2 is asymptotically sharp.

Remark 1.4. The assumption that C is $\mathbb{F}_q$-irreducible cannot be dropped. To see this, fix an integer $\ell \geq 2$ and, for $q \geq \ell$, choose distinct elements $c_1, \ldots, c_\ell \in \mathbb{F}_q$. Let C be the union of the ℓ $\mathbb{F}_q$-conics obtained as the projective closures of

$$y = x^2 + c_i, \qquad 1 \leq i \leq \ell.$$

The sets of affine $\mathbb{F}_q$-points of these conics are pairwise disjoint, while all of the conics contain the point $[0 : 1 : 0]$ at infinity. Consequently,

$$|C(\mathbb{F}_q)| = \ell q + 1.$$

On the other hand, any line meets each of the ℓ conics in at most two points and therefore contains at most 2ℓ points of $C(\mathbb{F}_q)$. It follows that

$$\operatorname{cov}(C) \geq \left\lceil \frac{\ell q + 1}{2\ell} \right\rceil > \frac{q}{2}.$$

Since $\deg(C) = 2\ell$, the bound of Theorem 1.2 fails for C whenever $0 < \varepsilon < \frac{1}{2} - \frac{1}{2\ell}$.

The proof of Theorem 1.2 in the geometrically irreducible case has two main steps. First, we show that, outside a bounded exceptional set, the number of $\mathbb{F}_q$-lines through a smooth point $P \in C(\mathbb{F}_q)$ that meet C in d distinct smooth $\mathbb{F}_q$-points is

$$\frac{q}{(d-1)!} + O_d(\sqrt{q}).$$

This estimate is obtained by studying the monodromy of projection from P and applying Entin's Chebotarev density theorem. We then form a d-uniform hypergraph whose vertices are the smooth $\mathbb{F}_q$-points of C outside this exceptional set and whose edges are the sets of d collinear vertices. The preceding estimate shows that this hypergraph is nearly regular, and the Pippenger–Spencer theorem produces the desired cover.

The intersection of lines and plane curves over finite fields is a recurring theme in arithmetic geometry and incidence combinatorics. Much of the recent literature concerns completely split line sections, which intersect the curve in the maximum possible number of distinct rational points. The existence of these lines also plays a key role in the proof of our main theorem. In related work, Diem [Die12] counts line-section divisors that split completely into $\mathbb{F}_q$-points on reflexive curves, and his [Die12, Proposition 10] closely parallels our Proposition 2.5. From a different perspective, Makhul, Schicho, and Gallet [MSG20] use a function-field Chebotarev theorem to analyze the distribution of rational points within these line sections. While these earlier studies focus on counting or local distributions, we solve a global covering problem.

We conclude the introduction with a second motivation for the problem. When $d = 3$ and C is smooth with an $\mathbb{F}_q$-rational flex O, choosing O as the identity makes C an elliptic curve with the usual chord–tangent group law. In this case we can draw on existing results about abelian groups, in particular on the literature concerning the "zero-sum property."

For such an elliptic curve $E/\mathbb{F}_q$, we have $E(\mathbb{F}_q) \cong \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$ with $a \mid b$ [Was08, Theorem 4.1, p. 97]. A partition of $E(\mathbb{F}_q)$ into zero-sum triples yields a cover by the corresponding lines, since a line meets E in three points summing to O under the chord–tangent law. Writing $n = |E(\mathbb{F}_q)|$, if n is odd, Tannenbaum's theorem [Tan81] on Skolem partitions gives

$$\operatorname{cov}(E) = \frac{n}{3} + O(1).$$

However, for $d > 3$ the set $C(\mathbb{F}_q)$ has no analogous group structure, so we instead turn to tools from arithmetic geometry and the probabilistic method.

2. Monodromy of projections from points

In the geometrically irreducible case, the first step in the proof of Theorem 1.2 is to estimate the number of $\mathbb{F}_q$-lines through a smooth point $P \in C(\mathbb{F}_q)$ that meet C in d distinct smooth $\mathbb{F}_q$-points. We do so by studying the projection from P.

This projection gives a finite cover of the pencil of lines through P, whose fiber over a line consists of the remaining $d-1$ points of intersection with C. All of these points are defined over $\mathbb{F}_q$ exactly when the Frobenius map acts trivially on the fiber. Entin's Chebotarev density theorem [Ent21] counts the fibers with this property once the monodromy group of the cover is known. We begin by recalling the necessary terminology.

Let $\phi\colon X \to Y$ be a finite étale morphism of degree m between quasiprojective varieties defined over $\mathbb{F}_q$, where Y is normal and geometrically irreducible. Fix a geometric point of Y; since ϕ is finite étale, the fiber over this point consists of m distinct points. The étale fundamental group of Y permutes these points, and choosing an ordering of the fiber turns this action into a homomorphism to the symmetric group S_m. A different ordering conjugates the image, so the resulting subgroup of S_m is well defined up to conjugacy. We call this subgroup the *arithmetic monodromy group* of ϕ and denote it by $\mathrm{Mon}(X/Y)$.

After extending the base field from $\mathbb{F}_q$ to $\overline{\mathbb{F}_q}$, the same construction gives the *geometric monodromy group*

$$\mathrm{Mon}^{\mathrm{g}}(X/Y) = \mathrm{Mon}\big(X \times_{\mathbb{F}_q} \overline{\mathbb{F}_q} \,/\, Y \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}\big).$$

The geometric monodromy group is a normal subgroup of the arithmetic monodromy group. More precisely, there is an integer $\nu \geq 1$ and an exact sequence

$$1 \longrightarrow \mathrm{Mon}^{\mathrm{g}}(X/Y) \longrightarrow \mathrm{Mon}(X/Y) \longrightarrow \mathrm{Gal}(\mathbb{F}_{q^\nu}/\mathbb{F}_q) \longrightarrow 1, \tag{2}$$

where $\mathbb{F}_{q^\nu}$ is the smallest finite extension of $\mathbb{F}_q$ over which the arithmetic monodromy group of the base-changed cover coincides with the geometric monodromy group.

For a point $y \in Y(\mathbb{F}_q)$, the q-power Frobenius map permutes the m geometric points of the fiber $\phi^{-1}(y)$. We call the conjugacy class of the resulting permutation in $\mathrm{Mon}(X/Y)$ the *Frobenius class* of ϕ at y and denote it by $\mathrm{Fr}_\phi(y)$. Its cycle lengths are the degrees of the closed points in the fiber. In particular, $\mathrm{Fr}_\phi(y)$ is the identity class precisely when the fiber consists of m distinct $\mathbb{F}_q$-points. The group $\mathrm{Gal}(\mathbb{F}_{q^\nu}/\mathbb{F}_q)$ is cyclic, generated by the automorphism $x \mapsto x^q$, and the final map in (2) sends every Frobenius class $\mathrm{Fr}_\phi(y)$ to this generator.

The same definitions and hypotheses apply over any field F, with arithmetic monodromy over F and geometric monodromy over an algebraic closure $\overline{F}$. For a finite separable map from a smooth projective integral curve to $\mathbb{P}^1_F$, monodromy is the Galois group of the Galois closure of the extension of function fields. It stays the same after restricting to a nonempty open subset over which the map is étale.

The implied constant in Entin's error term depends only on the complexity of the cover, which we now define. A locally closed set $Z \subset \mathbb{P}^N$ has *complexity at most* M if $N \leq M$ and Z can be described by equations $F_1 = \cdots = F_s = 0$ together with a single nonvanishing condition $G \neq 0$, where $F_1, \ldots, F_s$ and G are homogeneous polynomials of degree at most M. The complexity of a morphism ϕ is the complexity of its graph, embedded in projective space via the Segre embedding. We write $\mathrm{comp}(\phi)$ for the complexity of ϕ, and $O_{\mathrm{comp}(\phi)}(\cdot)$ denotes a bound whose implied constant depends only on $\mathrm{comp}(\phi)$. In particular, a bound on the complexity gives an estimate uniform in q.

We can now state the form of Entin's Chebotarev density theorem [Ent21, Theorem 3] that we need. It counts the points $y \in Y(\mathbb{F}_q)$ whose Frobenius class is a prescribed conjugacy class $\mathcal{C}$, under the necessary condition that $\mathcal{C}$ maps to the Frobenius generator of $\mathrm{Gal}(\mathbb{F}_{q^\nu}/\mathbb{F}_q)$.

Theorem 2.1 (Entin's Chebotarev density theorem)**.** *Let $\phi\colon X \to Y$ be a finite étale morphism of quasiprojective varieties defined over $\mathbb{F}_q$, where Y is normal and geometrically irreducible. Let $\mathcal{C}$ be a conjugacy class in $\mathrm{Mon}(X/Y)$ whose image under the final map in (2) is the automorphism $x \mapsto x^q$. Then*

$$\big|\{y \in Y(\mathbb{F}_q) : \mathrm{Fr}_\phi(y) = \mathcal{C}\}\big| = \frac{\nu|\mathcal{C}|}{|\,\mathrm{Mon}(X/Y)|} q^{\dim Y}\left(1 + O_{\mathrm{comp}(\phi)}(q^{-1/2})\right).$$

Here $\nu \geq 1$ is the integer appearing in (2).

When Y is a curve, so that $\dim Y = 1$, the error term is $O_{\mathrm{comp}(\phi)}(\sqrt{q})$.

We shall apply Theorem 2.1 to the following cover. For a smooth geometric point $P \in C$, let $\mathcal{L}_P \cong \mathbb{P}^1$ be the pencil of lines through P. Let $U_P \subset \mathcal{L}_P$ be the open set of those lines L for which $L \cap C$ consists of d distinct smooth geometric points, and define

$$X_P = \{(Q, L) \in C_{\mathrm{sm}} \times U_P : Q \in L,\ Q \neq P\}.$$

Consider the projection

$$\phi_P \colon X_P \longrightarrow U_P, \qquad (Q, L) \longmapsto L.$$

For every $L \in U_P$, the fiber $\phi_P^{-1}(L)$ consists of the $d-1$ points of $(L \cap C) \setminus \{P\}$.

To count the completely split fibers, we will apply Theorem 2.1 with the identity conjugacy class. We first establish that ϕ_P is finite étale in Lemma 2.2. Then Lemma 2.3 computes its geometric monodromy when P is the generic point of C, and Lemma 2.4 gives the geometric and arithmetic monodromy groups outside a set of points whose size is bounded in terms of d alone.

Lemma 2.2. *Let $C \subset \mathbb{P}^2$ be a geometrically irreducible reflexive curve of degree $d \geq 2$ defined over $\mathbb{F}_q$, and put $k = \overline{\mathbb{F}_q}$. For every $P \in C_{\mathrm{sm}}(k)$, the set U_P is nonempty and the map $\phi_P \colon X_P \to U_P$ is finite étale of degree $d-1$.*

Proof. We work over k and fix $P \in C_{\mathrm{sm}}(k)$. Let $\widetilde{C}$ be the normalization of C, and let $\widetilde{P}$ be the unique point above P. The pencil $\mathcal{L}_P$ cuts out a linear series on $\widetilde{C}$ with base point $\widetilde{P}$. Removing this base point leaves a base-point-free linear series and defines the residual map

$$\pi_P \colon \widetilde{C} \longrightarrow \mathcal{L}_P$$

of degree $d-1$. Concretely, if $Q \neq P$ is a smooth point of C and $\widetilde{Q}$ is its lift to $\widetilde{C}$, then $\pi_P(\widetilde{Q})$ is the line PQ, while $\pi_P(\widetilde{P})$ is the tangent line $T_P C$.

We first check that π_P is separable. A point $R \in \mathbb{P}^2$ is called a *strange point* for C if every tangent line to C at a smooth point passes through R. The curve C is called *strange* if it has a strange point. A reflexive curve of degree at least two has no strange point [Hef89, p. 4], and projection from a point R is separable on C unless R is a strange point; see the discussion before [Ent21, Lemma 2.2]. In particular, projection from P is separable. The map π_P induces the same extension of function fields, so it is separable as well.

Being separable, π_P has only finitely many branch values. A line in $\mathcal{L}_P$ lies in U_P exactly when it is not a branch value, is not the tangent line $T_P C$, and does not pass through a singular point of C. Only finitely many lines are excluded by these conditions, so U_P is nonempty. Since U_P contains no branch values, the restriction of π_P over U_P is finite étale of degree $d-1$, and the normalization map identifies it with $\phi_P \colon X_P \to U_P$. □

The maps ϕ_P and π_P induce the same extension of function fields, so they have the same monodromy group. We now determine this group at the generic point of C.

Lemma 2.3. *Let C and k be as in Lemma 2.2. After base change to k, let η be the generic point of C and set $K = k(C)$. Viewing η as a smooth K-rational point of $C \times_k K$, define $\phi_\eta \colon X_\eta \to U_\eta$ by the same construction. Its monodromy group is S_{d-1}, both over K and over an algebraic closure $\overline{K}$.*

Proof. We keep the notation of the proof of Lemma 2.2. The same proof over K, with η in place of P, gives a separable residual map π_η and a nonempty open set U_η over which its restriction ϕ_η is finite étale of degree $d-1$. If $d = 2$, then π_η has degree one and trivial monodromy over both K and $\overline{K}$. We may assume from now on that $d \geq 3$. We first compute the monodromy group over K, and then show that it is the same over $\overline{K}$.

Let $\Omega \subset (\mathbb{P}^2)^\vee$ be the open set of lines that meet C in d distinct smooth points, and let V be the surface of pairs (P, L) with $L \in \Omega$ and $P \in L \cap C$. The projection $V \to \Omega$ is finite étale of degree d, and its monodromy group is S_d by results of Rathmann [Rat87] and of Ballico and Hefez [BH86]; see [Ent21, Theorem 2].

To describe the remaining intersection points after P has been marked, consider the cover $V' \to V$ of triples (P, Q, L) with $(P, L) \in V$ and $Q \in (L \cap C) \setminus \{P\}$. This cover has degree $d-1$. Since V marks one of the d intersection points, the monodromy group of $V' \to V$ is the stabilizer S_{d-1} of that point in S_d, acting on the remaining $d-1$ points.

To relate this cover to ϕ_η, we take generic fibers of the projection $V \to C_{\mathrm{sm}}$, $(P, L) \mapsto P$. Its generic fiber is U_η, a nonempty open subset of the pencil $\mathcal{L}_\eta \cong \mathbb{P}^1_K$. Over this fiber, the cover $V' \to V$ parametrizes pairs (Q, L) with $L \in U_\eta$ and $Q \in (L \cap C) \setminus \{\eta\}$, and sends (Q, L) to L. This is precisely $\phi_\eta \colon X_\eta \to U_\eta$. Passing to the generic fibers does not change the extension of function fields, so

$k(V')/k(V)$ is the extension induced by ϕ_η, and hence also by π_η. It follows that the monodromy group of π_η over K is S_{d-1}.

Since K is not algebraically closed, this group may become smaller after base change to $\overline{K}$. Write $\bar{\eta}$ for the resulting $\overline{K}$-point and $\pi_{\bar{\eta}}$ for the base-changed map, and let H be its monodromy group. The geometric monodromy group is normal in the monodromy group over K, so $H \trianglelefteq S_{d-1}$. We will now show that $H = S_{d-1}$. The transpositions form a single conjugacy class of S_{d-1} and generate it, so it suffices to find one transposition in H.

Since C is reflexive, the characteristic is not two, and a general tangent line has contact order two at one smooth point and meets C transversely at $d-2$ further smooth points [Ent21, Section 2]. Call such a line a *simple tangent*. To obtain the desired transposition, we seek a simple tangent that meets C transversely at $\bar{\eta}$.

Let I be the variety of pairs (R, L) where L is a simple tangent and R is a point where L meets C transversely. Let $\Theta \subset C$ be the image of $I \to C$, $(R, L) \mapsto R$. This image is constructible, and we claim that it is dense. If not, then Θ is finite. Since $d \geq 3$, every simple tangent passes through a point of Θ. For each $R \in \Theta$, the set $\{L \in C^\vee : R \in L\}$ is closed in $C^\vee$. These finitely many closed subsets together contain all simple tangents, which form a dense subset of the irreducible curve $C^\vee$, so one of them is all of $C^\vee$. The corresponding point R would then be a strange point of C. This is impossible for a reflexive curve, as recalled in the proof of Lemma 2.2.

The projection $I \to C$ is therefore dominant, so its generic fiber is nonempty. Since this fiber is of finite type over K, it has a $\overline{K}$-point, giving a simple tangent L_0 that meets C transversely at $\bar{\eta}$. The fiber of $\pi_{\bar{\eta}}$ over L_0 consists of the point of tangency, with ramification index two, and $d-3$ unramified points. This ramification is tame because $\mathrm{char}(k) \neq 2$, and it yields a transposition in H, which forces $H = S_{d-1}$. The same holds for ϕ_η, since restricting π_η to U_η does not change its monodromy group. $\square$

The next lemma carries this conclusion from the generic point to every smooth point of C outside a finite set, whose size we bound in terms of the degree.

Lemma 2.4. *Let $C \subset \mathbb{P}^2$ be a geometrically irreducible reflexive curve of degree $d \geq 2$ defined over $\mathbb{F}_q$. There exists a set $B_C \subset C_{\mathrm{sm}}(\overline{\mathbb{F}_q})$, with $|B_C| = O_d(1)$, such that the following holds for every $P \in C_{\mathrm{sm}}(\overline{\mathbb{F}_q}) \setminus B_C$.*

The map $\phi_P \colon X_P \to U_P$ is finite étale of degree $d-1$, and its geometric monodromy group is S_{d-1}. If $P \in C(\mathbb{F}_q)$, then the arithmetic monodromy group of ϕ_P is also S_{d-1}. Moreover, the complexity of ϕ_P is $O_d(1)$, uniformly in C, q, and P.

Proof. Work over $k = \overline{\mathbb{F}_q}$ and put $K = k(C)$. We first show that the monodromy group of ϕ_P is S_{d-1} for all but finitely many $P \in C_{\mathrm{sm}}(k)$. We express this condition as the irreducibility of an auxiliary cover, because geometric irreducibility passes from the generic fiber to the fibers over an open set.

For $P \in C_{\mathrm{sm}}(k)$, let $W_P \to U_P$ be the cover obtained by ordering the $d-1$ points in each fiber of the finite étale map ϕ_P from Lemma 2.2. The monodromy group G_P of ϕ_P acts freely on the $(d-1)!$ orderings of a fiber, and the irreducible components of W_P correspond to the $(d-1)!/|G_P|$ orbits. In particular, W_P is irreducible exactly when $G_P = S_{d-1}$.

Let W be the variety of tuples $(P, Q_1, \ldots, Q_{d-1}, L)$ in which L meets C in the d distinct smooth points $P, Q_1, \ldots, Q_{d-1}$. The fiber of the projection

$$W \longrightarrow C_{\mathrm{sm}}, \qquad (P, Q_1, \ldots, Q_{d-1}, L) \longmapsto P$$

over P is W_P, and its generic fiber is the ordered cover W_η attached to ϕ_η. By Lemma 2.3 and the criterion above, applied over $\overline{K}$, the generic fiber W_η is geometrically irreducible. This property extends to the fibers over a dense open subset of C_{sm} [Sta26, Tag 0559]. The complement of a dense open subset of a curve is finite, so the set

$$B_C = \{P \in C_{\mathrm{sm}}(k) : W_P \text{ is not irreducible}\}$$

is finite. By the criterion above, the monodromy group of ϕ_P is S_{d-1} for every $P \in C_{\mathrm{sm}}(k) \setminus B_C$. When $d = 2$, the cover $W_P \to U_P$ has degree one for every P, so $B_C = \varnothing$.

It remains to bound $|B_C|$ in terms of d alone, and for this we let the curve vary. Let $\mathcal{P}$ be the projective space over $\mathbb{Z}$ parametrizing plane curves of degree d, and let $\mathcal{U} \to \mathcal{P}$ be the universal family of pairs (C, P) with P smooth on C. The construction of W gives a family of finite type over $\mathcal{U}$ whose fiber over (C, P) is W_P. Let $\mathcal{E} \subset \mathcal{U}$ be the set of points over which this fiber is not geometrically irreducible. This set is constructible [Sta26, Tag 055B], and for a curve C as in the lemma its geometric fiber over $[C] \in \mathcal{P}$ is B_C.

Write $\mathcal{E}$ as a finite union of locally closed subschemes $\mathcal{E}_1, \dots, \mathcal{E}_s$. For each i, the set of points at which $\mathcal{E}_i \to \mathcal{P}$ is quasi-finite is open in $\mathcal{E}_i$ [Sta26, Tag 01TI]. For a curve C as in the lemma, the geometric fiber of each $\mathcal{E}_i$ over $[C]$ is a subset of the finite set B_C. Each point is isolated in that fiber and therefore lies in the quasi-finite locus. This open locus is quasi-compact because $\mathcal{E}_i$ is Noetherian, so there is an integer N_i bounding the number of points in each of its geometric fibers over $\mathcal{P}$ [Sta26, Tag 03JA]. We obtain $|B_C| \le N_1 + \cdots + N_s$. Since the whole construction takes place over $\mathbb{Z}$ for fixed d, the integers s and N_i depend only on d, which gives $|B_C| = O_d(1)$.

Now suppose that $P \in C_{\mathrm{sm}}(\mathbb{F}_q) \setminus B_C$. Then ϕ_P is defined over $\mathbb{F}_q$, and its arithmetic monodromy group is a subgroup of S_{d-1} that contains the geometric monodromy group. We have just seen that the latter is all of S_{d-1}, so the two groups coincide.

It remains to bound the complexity of ϕ_P, and here P may be any smooth point of C. Choose projective coordinates with $P = [0 : 1 : 0]$ and $T_P C = \{z = 0\}$. If $F = 0$ defines C, these coordinate choices force the coefficients of y^d and xy^{d-1} in F to vanish, while the coefficient of zy^{d-1} is nonzero. After scaling F, the polynomial $f(x, y) = F(x, y, 1)$ is therefore monic of degree $d - 1$ in y.

Let $D(x) = \mathrm{Disc}_y(f)(x)$ be the discriminant of f with respect to y. The lines of the pencil other than $T_P C$ have equations $x = az$. Such a line lies in U_P exactly when $f(a, y)$ has $d - 1$ distinct roots, or equivalently when $D(a) \neq 0$; at each root, $f_y \neq 0$, so the corresponding point of C is smooth. In these coordinates, ϕ_P is the projection $(x, y) \mapsto x$ from $\{f = 0,\ D \neq 0\}$ to $\{D \neq 0\}$. The degrees of f and D are bounded in terms of d, so the graph of ϕ_P has complexity $O_d(1)$, uniformly in C, q, and P. □

We can now apply Entin's theorem to count the completely split fibers of ϕ_P. These fibers correspond precisely to the $\mathbb{F}_q$-lines through P that meet C in d distinct smooth $\mathbb{F}_q$-points.

Proposition 2.5. *Let C and B_C be as in Lemma 2.4, and let $P \in C_{\mathrm{sm}}(\mathbb{F}_q) \setminus B_C$. The number of $\mathbb{F}_q$-lines through P that meet C in d distinct smooth $\mathbb{F}_q$-points is*

$$\frac{q}{(d-1)!} + O_d(\sqrt{q}).$$

The implied constant is uniform in C, q, and P.

Proof. Apply Theorem 2.1 to the finite étale morphism $\phi_P \colon X_P \to U_P$. By Lemma 2.4, both the geometric and arithmetic monodromy groups of ϕ_P are S_{d-1}. The exact sequence (2) then gives $\nu = 1$, and the identity conjugacy class satisfies the hypothesis of Entin's theorem.

A line $L \in U_P(\mathbb{F}_q)$ has Frobenius class equal to the identity precisely when every point of the fiber $\phi_P^{-1}(L)$ is defined over $\mathbb{F}_q$. Since this fiber is $(L \cap C) \setminus \{P\}$, that happens exactly when L meets C in d distinct smooth $\mathbb{F}_q$-points.

The identity conjugacy class has size one, while $|S_{d-1}| = (d-1)!$. Since U_P is a curve and the complexity of ϕ_P is $O_d(1)$, Theorem 2.1 gives

$$\frac{q}{(d-1)!} + O_d(\sqrt{q})$$

such $\mathbb{F}_q$-lines. The complexity bound is uniform in C, q, and P, so the same is true of the implied constant. □

A closely related estimate is stated in [Die12, Proposition 10], under the hypothesis that only ordinary tangents pass through P. For singular plane models, projection may ramify at a branch even when the corresponding line is not tangent to that branch. We have given an independent argument in the proofs of Lemmas 2.3 and 2.4 that accounts for this possibility and yields the uniform estimate used in Section 3.

3. Proof of Theorem 1.2

We first dispose of the case in which C is not geometrically irreducible. As observed in the introduction, in this case

$$|C(\mathbb{F}_q)| \le \frac{d^2}{4}.$$

These points can be covered by at most $d^2/4$ $\mathbb{F}_q$-lines, and $d^2/4 \le \left(\frac{1}{d} + \varepsilon\right) q$ after enlarging $M_{d,\varepsilon}$ if necessary. We may therefore assume that C is geometrically irreducible. By the hypothesis of Theorem 1.2, the curve C is then reflexive.

We now complete the proof by combining the estimate from Proposition 2.5 with the Pippenger–Spencer theorem [PS89] on covers of nearly regular hypergraphs.

A *cover* of a hypergraph is a set of edges whose union contains all vertices. Here $d(v)$ denotes the degree of a vertex v, and $d(v,w)$ the number of edges containing both v and w. We use the following form of the Pippenger–Spencer theorem, taken from Alon and Spencer [AS16, Theorem 4.7.1].

Theorem 3.1. *For every integer $r \geq 2$ and all reals $k \geq 1$ and $a > 0$, there are $\gamma = \gamma(r,k,a) > 0$ and $d_0 = d_0(r,k,a)$ such that for every $n \in \mathbb{N}$ and $D \geq d_0$ the following holds.*

Let $H = (V,E)$ be an r-uniform hypergraph on n vertices in which every vertex has positive degree. Suppose that

1. *all but at most γn vertices $v \in V$ satisfy $(1-\gamma)D \leq d(v) \leq (1+\gamma)D$;*
2. *every vertex $v \in V$ satisfies $d(v) < kD$;*
3. *every pair of distinct vertices $v, w \in V$ satisfies $d(v,w) < \gamma D$.*

Then H has a cover consisting of at most $(1+a)n/r$ edges.

Let $S \subset C(\mathbb{F}_q)$ be the set of singular $\mathbb{F}_q$-points, and let B_C be the exceptional set from Lemma 2.4. Both $|S|$ and $|B_C|$ are bounded in terms of d, so the set

$$T = S \cup \big(B_C \cap C(\mathbb{F}_q)\big)$$

has size $O_d(1)$. We first cover the points of T: each lies on some $\mathbb{F}_q$-line, so at most $|T| = O_d(1)$ lines suffice. It remains to cover the points of $C(\mathbb{F}_q) \setminus T$.

To apply Theorem 3.1, we describe the hypergraph $H = (V,E)$. The vertex set is $V = C(\mathbb{F}_q) \setminus T$, and the edges are the sets of d distinct collinear points of V. Equivalently, each edge arises from an $\mathbb{F}_q$-line meeting C in d distinct smooth $\mathbb{F}_q$-points, none of which lies in T. Thus, H is a d-uniform hypergraph.

We compute the degree of a vertex $v \in V$, which corresponds to a point $P \in C(\mathbb{F}_q) \setminus T$. By Proposition 2.5, the number of $\mathbb{F}_q$-lines through P that meet C in d distinct smooth $\mathbb{F}_q$-points is

$$\frac{q}{(d-1)!} + O_d(\sqrt{q}).$$

Among these lines, at most $|T| = O_d(1)$ of them contain a point of T, since each point of T determines a unique line through P. After these lines are discarded, the remaining lines meet C in d distinct smooth $\mathbb{F}_q$-points of V, and each resulting set of points is an edge of H containing v. We obtain

$$d(v) = \frac{q}{(d-1)!} + O_d(\sqrt{q}) \tag{3}$$

for every $v \in V$. In particular, every vertex has positive degree when q is sufficiently large.

We apply Theorem 3.1 with the following parameters. Fix an arbitrary $a > 0$, and set $r = d$ and $k = 2(d-1)!$. With these choices, $d_0(r,k,a)$ is a constant, so $D = \frac{q}{(d-1)!}$ exceeds d_0 once q is sufficiently large; we take this value of D. Finally, the Aubry–Perret bound [AP96] gives $|C(\mathbb{F}_q)| = q + O_d(\sqrt{q})$, so $n = |V| = |C(\mathbb{F}_q)| - |T| = q + O_d(\sqrt{q})$.

We now verify the three conditions. Here $\gamma = \gamma(r,k,a)$ and $d_0 = d_0(r,k,a)$ denote the constants supplied by Theorem 3.1.

(1) Using (3), we obtain

$$\frac{q}{(d-1)!}\left(1 - O_d\left(\frac{1}{\sqrt{q}}\right)\right) \leq d(v) \leq \frac{q}{(d-1)!}\left(1 + O_d\left(\frac{1}{\sqrt{q}}\right)\right).$$

Since $\gamma > 0$ is a fixed constant and $D = \frac{q}{(d-1)!}$, this gives $(1-\gamma)D \leq d(v) \leq (1+\gamma)D$ for sufficiently large q. This holds for every vertex of H, so the exceptional set allowed in condition (1) is in fact empty.

(2) Every point of $\mathbb{P}^2(\mathbb{F}_q)$ lies on exactly $q+1$ distinct $\mathbb{F}_q$-lines, so $d(v) \leq q+1$ for every v. Since

$$kD = 2(d-1)! \cdot \frac{q}{(d-1)!} = 2q,$$

we have $d(v) \leq q+1 < 2q = kD$ for every $v \in V$.

(3) Two distinct $\mathbb{F}_q$-points lie on a unique $\mathbb{F}_q$-line, and any edge containing both v and w consists of d collinear points on that line. Hence at most one edge contains both of them, so $d(v,w) \leq 1$. Since $\gamma D = \gamma \frac{q}{(d-1)!} \to \infty$, we have $d(v,w) < \gamma D$ for all sufficiently large q.

By Theorem 3.1, the hypergraph H has a cover consisting of at most

$$(1+a)\frac{n}{d}$$

edges. Since $n = |V| = q + O_d(\sqrt{q})$, there is a constant K_d such that $n \le q + K_d\sqrt{q}$ for all sufficiently large q. The number of edges in the cover is at most

$$(1+a)\frac{q + K_d\sqrt{q}}{d} = \frac{q}{d} + \frac{aq}{d} + \frac{(1+a)K_d}{d}\sqrt{q}.$$

Let R_d be a constant such that the points of T can be covered using at most R_d additional $\mathbb{F}_q$-lines. Recall that $a > 0$ was arbitrary. Given $\varepsilon > 0$, take a small enough that $a/d < \varepsilon/3$, and then q large enough that

$$\frac{(1+a)K_d}{d}\sqrt{q} < \frac{\varepsilon q}{3} \qquad \text{and} \qquad R_d < \frac{\varepsilon q}{3}.$$

Combining the hypergraph cover with the additional lines used to cover T, the total number of $\mathbb{F}_q$-lines is at most

$$\frac{q}{d} + \frac{\varepsilon q}{3} + \frac{\varepsilon q}{3} + \frac{\varepsilon q}{3} = \left(\frac{1}{d} + \varepsilon\right) q.$$

This proves Theorem 1.2.

4. An obstruction in every positive characteristic

In this section, we show that the reflexivity hypothesis in Theorem 1.2 cannot be omitted in any positive characteristic. For every prime p, we exhibit a geometrically irreducible nonreflexive curve of degree p^2, defined over $\mathbb{F}_p$, whose covering number over $\mathbb{F}_q$, for $q = p^{2n+1}$, is $q/p + 1$. These curves fall outside Theorem 1.2 because they are nonreflexive, and outside Corollary 1.3 because $p^2 > p$.

The obstruction is visible in the proof itself. The application of Entin's theorem in Section 2 depends on the identity conjugacy class satisfying the compatibility condition in Theorem 2.1. This condition holds exactly when $\nu = 1$, and in general one may have $\nu > 1$. The following example shows what can go wrong in every positive characteristic.

Proposition 4.1. *Let p be a prime, let $q = p^{2n+1}$, where $n \ge 0$, and let $C \subset \mathbb{P}^2$ be the curve of degree p^2 given by*

$$yz^{p^2-1} = x^{p^2}.$$

Then C is geometrically irreducible and nonreflexive, and $|C(\mathbb{F}_q)| = q + 1$, but every $\mathbb{F}_q$-line contains at most p points of $C(\mathbb{F}_q)$. Moreover,

$$\operatorname{cov}(C) = \left\lceil \frac{q+1}{p} \right\rceil = \frac{q}{p} + 1.$$

Proof. Regard $yz^{p^2-1} - x^{p^2}$ as a polynomial in y with coefficients in $\overline{\mathbb{F}_q}[x, z]$. It is linear in y, and its coefficients z^{p^2-1} and x^{p^2} are coprime, so it is irreducible in $\overline{\mathbb{F}_q}[x, y, z]$. This proves that C is geometrically irreducible. It is the projective closure of the affine rational curve

$$y = x^{p^2}.$$

The curve is parametrized by

$$[s : t] \longmapsto [st^{p^2-1} : s^{p^2} : t^{p^2}].$$

On the affine chart $t = 1$, this gives the points $[s : s^{p^2} : 1]$, while $t = 0$ gives the point $P_\infty = [0 : 1 : 0]$. These exhaust $C(\mathbb{F}_q)$, so

$$C(\mathbb{F}_q) = \{[s : s^{p^2} : 1] : s \in \mathbb{F}_q\} \cup \{P_\infty\},$$

and $|C(\mathbb{F}_q)| = q + 1$.

The point P_∞ is singular, whereas every affine point is smooth. At the point $[s : s^{p^2} : 1]$, the tangent line is $y = s^{p^2} z$. All these tangent lines pass through $[1 : 0 : 0]$. We conclude that C is a strange curve, and therefore nonreflexive.

Let L be an $\mathbb{F}_q$-line with equation $ax + by + cz = 0$. Suppose first that $b = 0$, so L has equation $ax + cz = 0$ with $(a, c) \ne (0, 0)$. Then P_∞ lies on L, and substituting $[s : s^{p^2} : 1]$ into the equation gives $as + c = 0$. This linear equation has at most one solution $s \in \mathbb{F}_q$, so L contains at most two points of $C(\mathbb{F}_q)$, hence at most p since $p \ge 2$.

Now suppose that $b \ne 0$. Then L does not contain P_∞, and its intersections with the affine part of C correspond to the roots in $\mathbb{F}_q$ of

$$bs^{p^2} + as + c = 0.$$

Set $A(T) = bT^{p^2} + aT$, an additive polynomial. If $A(r) = -c$, then the solutions of $A(s) = -c$ are exactly the elements of the coset $r + \ker(A)$.

It is enough to bound the size of $\ker(A)$. If $a = 0$, then $\ker(A) = \{0\}$. Suppose that $a \neq 0$. If $0 \neq u \in \ker(A)$, then

$$u^{p^2-1} = -\frac{a}{b}.$$

If this equation has a solution $u_0 \in \mathbb{F}_q^\times$, then all of its solutions are of the form $u_0 w$, where $w^{p^2-1} = 1$. Since

$$\gcd(p^2-1, q-1) = \gcd(p^2-1, p^{2n+1}-1) = p^{\gcd(2,2n+1)} - 1 = p - 1,$$

there are exactly $p-1$ solutions to the latter equation in $\mathbb{F}_q^\times$. Therefore, $\ker(A)$ contains at most $p-1$ nonzero elements, yielding $|\ker(A)| \leq p$. Since every nonempty fiber of A is a coset of $\ker(A)$, the equation $bs^{p^2} + as + c = 0$ has at most p roots in $\mathbb{F}_q$.

We have shown that every $\mathbb{F}_q$-line contains at most p points of $C(\mathbb{F}_q)$. Since $|C(\mathbb{F}_q)| = q+1$, covering these points requires at least

$$\left\lceil \frac{q+1}{p} \right\rceil = \frac{q}{p} + 1$$

lines. For the reverse inequality, consider the additive map

$$B \colon \mathbb{F}_q \longrightarrow \mathbb{F}_q, \qquad B(s) = s^{p^2} - s.$$

Its kernel is

$$\ker(B) = \mathbb{F}_{p^2} \cap \mathbb{F}_q = \mathbb{F}_p,$$

so $|\operatorname{im}(B)| = q/p$. For each $r \in \operatorname{im}(B)$, take the line

$$L_r \colon y = x + rz.$$

Every affine point $[s : s^{p^2} : 1]$ lies on $L_{B(s)}$. These q/p lines cover all affine points of $C(\mathbb{F}_q)$, and the line $z = 0$ covers $P_\infty = [0 : 1 : 0]$. It follows that

$$\operatorname{cov}(C) \leq \frac{q}{p} + 1.$$

Together with the lower bound established above, this proves

$$\operatorname{cov}(C) = \frac{q}{p} + 1,$$

as claimed. □

Since C has degree p^2 while

$$\operatorname{cov}(C) = \frac{q}{p} + 1,$$

these examples do not obey any bound of the form $\left(\frac{1}{p^2} + \varepsilon\right) q$ when $0 < \varepsilon < \frac{1}{p} - \frac{1}{p^2}$. This confirms that the reflexivity hypothesis in Theorem 1.2 is essential.

Remark 4.2. The preceding example also explains why the condition $\nu = 1$ in the application of Entin's theorem is not automatic. The smooth $\mathbb{F}_q$-points of C are

$$P_a = [a : a^{p^2} : 1], \qquad a \in \mathbb{F}_q.$$

Every line through P_a, other than the line $x = az$, can be written as

$$y - a^{p^2} z = m(x - az).$$

Substituting $[s : s^{p^2} : 1]$ into this equation gives

$$(s-a)^{p^2} = m(s-a).$$

Set $u = s - a$. The root $u = 0$ corresponds to the fixed point P_a. Removing this root and restricting to $m \neq 0$ leaves the equation

$$u^{p^2-1} = m$$

for the remaining points. Under the identification of U_{P_a} with $\mathbb{G}_m$, the residual cover ϕ_{P_a} is

$$\mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad u \longmapsto u^{p^2-1}.$$

Over $\overline{\mathbb{F}_q}$, this cover is Galois with geometric monodromy group C_{p^2-1}. All (p^2-1)-th roots of unity lie in $\mathbb{F}_{p^2}$, and a primitive one generates $\mathbb{F}_{p^2}$ over $\mathbb{F}_p$. Moreover, $\mathbb{F}_{p^2} \cap \mathbb{F}_q = \mathbb{F}_p$. Consequently, the smallest

extension of $\mathbb{F}_q$ containing all these roots of unity is $\mathbb{F}_{q^2}$. The Galois closure of $\mathbb{F}_q(u)/\mathbb{F}_q(m)$, where $m = u^{p^2-1}$, is $\mathbb{F}_{q^2}(u)$. The arithmetic monodromy group is $C_{p^2-1} \rtimes C_2$, where the nontrivial element of C_2 acts by

$$\zeta \longmapsto \zeta^q = \zeta^p.$$

Here we used $q = p^{2n+1} \equiv p \pmod{p^2-1}$. For $p = 2$, this specializes to $C_3 \rtimes C_2 \cong S_3$. In general, the exact sequence (2) becomes

$$1 \longrightarrow C_{p^2-1} \longrightarrow C_{p^2-1} \rtimes C_2 \longrightarrow \mathrm{Gal}(\mathbb{F}_{q^2}/\mathbb{F}_q) \longrightarrow 1.$$

In particular, $\nu = 2$, not 1.

For a parameter $m \in \mathbb{F}_q^\times$, the Frobenius class of the fiber over m maps to the nontrivial element of $\mathrm{Gal}(\mathbb{F}_{q^2}/\mathbb{F}_q)$, so it lies in the nontrivial coset of C_{p^2-1} in the arithmetic monodromy group. The identity, however, lies in C_{p^2-1} itself, so it cannot occur as the Frobenius class of any $\mathbb{F}_q$-fiber. Equivalently, the residual cover has no completely split fiber over $\mathbb{F}_q$. This is exactly the compatibility condition in Theorem 2.1: the identity maps to the trivial element of $\mathrm{Gal}(\mathbb{F}_{q^\nu}/\mathbb{F}_q)$, which is the Frobenius generator only when $\nu = 1$. By Lemma 2.4, for the covers ϕ_P used in the proof of Theorem 1.2, the geometric and arithmetic monodromy groups are both S_{d-1}. As a result, their quotient is trivial, so $\nu = 1$ and the identity class is allowed.

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